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COMMON HALFYEARLY EXAMINATION DECEMBER 2025

Standard-12

Reg No | | | | | | |

PART-III-MATHEMATICS

Time Allowed:3 Hours **Maximum Marks: 90**

Instructions: 1. Check the question paper for fairness of printing. If there is any Lack of fairness,inform the Hall Supervisor Immediately
2.Use Blue or Black Ink to write and underline and pencil to draw Diagrams.

PART-I

20x1=20

Note: **I) Answer all the questions.**
II) Choose the most appropriate answer from the given four alternatives And write the option code and the corresponding answer.

1. If A is an orthogonal matrix then
(1) $|A| = \pm 2$ (2) $|A| = 0$ (3) $|A| = \pm 1$ (4) $|A| = \pm 3$

2. If A, B and C are invertible matrices of some order, then which one of the following is not true?
(1) $\text{adj}\, A = |A| A^{-1}$ (2) $\text{adj}(AB) = (\text{adj}\, A)(\text{adj}\, B)$ (3) $\det A^{-1} = (\det A)^{-1}$ (4) $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$

3. The principal argument of $(\sin 40^\circ + i\cos 40^\circ)^5$ is
(1) -110° (2) -70° (3) 70° (4) 110°

4. If $x + iy = \left(-1 + i\sqrt{3}\right)^{2019}$, then x is
(1) 2^{2019} (2) -2^{2019} (3) -1 (4) 1

5. The polynomial $x^3 - kx^2 + 9x$ has three real zeros if and only if, k satisfies
(1) $|k| \leq 6$ (2) $k = 0$ (3) $|k| > 6$ (4) $|k| \geq 6$

6. If $\sin^{-1}\frac{x}{5} + \text{cosec}^{-1}\frac{5}{4} = \frac{\pi}{2}$, then the value of x is
(1) 4 (2) 5 (3) 2 (4) 3

7. The radius of the circle passing through the foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ and having its centre at $(0,3)$ is
(1) 3 (2) 4 (3) 5 (4) $\sqrt{7}$

8. If $x + y = k$ is a normal to the parabola $y^2 = 12x$, then the value of k is
(1) 3 (2) -1 (3) 1 (4) 9

9. If the straight line $\frac{x-3}{4} = \frac{y-4}{7} = \frac{z+3}{-13}$ lies in the plane $5x - y + z = p$, then the value of p is
(1) 2 (2) -3 (3) 8 (4) 9

10. If the direction cosines of a line are $\frac{1}{c}, \frac{1}{c}, \frac{1}{c}$, then
(1) $c = \pm 3$ (2) $c = \pm\sqrt{3}$ (3) $c > 0$ (4) $0 < c < 1$

11. The tangent to the curve $y^2 - xy + 9 = 0$ is vertical when
(1) $y = 0$ (2) $y = \pm\sqrt{3}$ (3) $y = \frac{1}{2}$ (4) $y = \pm 3$

12. One of the closest points on the curve $x^2 - y^2 = 4$ to the point $(6,0)$ is
(1) $(2,0)$ (2) $\left(\sqrt{5},1\right)$ (3) $\left(3,\sqrt{5}\right)$ (4) $\left(\sqrt{13},-\sqrt{3}\right)$

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13. If $w(x,y)=x^r, x>0$, then $\frac{\partial w}{\partial x}$ is equal to

(1) $x^r \log x$ (2) $y \log x$ (3) yx^{r-1} (4) $x \log y$

14. If $\frac{\Gamma(n+2)}{\Gamma(n)}=90$ then n is

(1) 10 (2) 5 (3) 8 (4) 9

15. The solution of $\frac{dy}{dx}+p(x)y=0$ is

(1) $y=ce^{\int pdx}$ (2) $y=ce^{-\int pdx}$ (3) $x=ce^{-\int pdy}$ (4) $x=ce^{\int pdy}$

16. The solution of $\frac{dy}{dx}=2^{y-x}$ is

(1) $2^x+2^y=C$ (2) $2^x-2^y=C$ (3) $\frac{1}{2^x}-\frac{1}{2^y}=C$ (4) $x+y=C$

17. If $P(X=0)=1-P(X=1)$. If $E(X)=3\text{Var}(X)$, then $P(X=0)$ is

(1) $\frac{2}{3}$ (2) $\frac{2}{5}$ (3) $\frac{1}{5}$ (4) $\frac{1}{3}$

18. If the function $f(x)=\frac{1}{12}$ for $a<x<b$, represents a probability density function of a continuous random variable X, then which of the following cannot be the value of a and b?

(1) 0 and 12 (2) 5 and 17 (3) 7 and 19 (4) 16 and 24

19. The operation $*$ defined by $a*b=\frac{ab}{7}$ is not a binary operation on

(1) $\mathbb{Q}^+$ (2) $\mathbb{Z}$ (3) $\mathbb{R}$ (4) $\mathbb{C}$

20. In the last column of the truth table for $\neg(p \vee \neg q)$ the number of final outcomes of the truth value 'F' are

(1) 1 (2) 2 (3) 3 (4) 4

PART-II

Answer any seven questions. Question no. 30 is compulsory 7x2=14

21. Obtain the Cartesian form of the locus of $z=x+iy$ in $\bar{z}=z^{-1}$.

22. Is $\cos^{-1}(-x)=\pi-\cos^{-1}(x)$ true? Justify your answer.

23. If the equation $3x^2+(3-p)xy+qy^2-2px=8pq$ represents a circle, find p and q. Also determine the centre and radius of the circle.

24. Prove that $[\vec{a}-\vec{b}, \vec{b}-\vec{c}, \vec{c}-\vec{a}]=0$.

25. Suppose $f(x)$ is a differentiable function for all x with $f'(x)\le 29$ and $f(2)=17$. What is the maximum value of $f(7)$?

26. Show that $F(x,y)=\frac{x^2+5xy-10y^2}{3x+7y}$ is a homogeneous function of degree 1.

27. Determine the order and degree (if exists) of the differential equations: $\frac{d^2y}{dx^2}+5\frac{dy}{dx}+\int y dx=x^3$

28. Compute $P(X=k)$ for the binomial distribution, $B(n,p)$ where $n=6, p=\frac{1}{3}, k=3$

29. Let $A=\{a+\sqrt{5}b : a,b \in \mathbb{Z}\}$. Check whether the usual multiplication is a binary operation on A.

30. If $A=\begin{bmatrix}3 & 4 & 5\\ -6 & 2 & -3\\ 8 & 1 & 7\end{bmatrix}$ then $|A^{-1}|=$

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PART-III

Answer any seven questions. Question no. 40 is compulsory 7x3=21

31. If $A=\begin{bmatrix} 8 & -4 \\ -5 & 3 \end{bmatrix}$, verify that $A(\text{adj } A)=(\text{adj } A)A=|A|I_2$.

32. Show that the equation $z^3+2\bar{z}=0$ has five solutions.

33. Show that the equation $x^9-5x^5+4x^4+2x^2+1=0$ has atleast 6 imaginary solutions.

34. If the normal at the point 't_1' on the parabola $y^2=4ax$ meets the parabola again at the point 't_2', then prove that $t_2=-\left(t_1+\frac{2}{t_1}\right)$.

35. Find the acute angle between the lines. $2x=3y=-z$ and $6x=-y=-4z$.

36. If $\lim_{\theta\to 0}\left(\frac{1-\cos m\theta}{1-\cos n\theta}\right)=1$, then prove that $m=\pm n$.

37. If $U(x,y,z)=\log(x^3+y^3+z^3)$, find $\frac{\partial U}{\partial x}+\frac{\partial U}{\partial y}+\frac{\partial U}{\partial z}$.

38. Evaluate $\int_2^3 \frac{\sqrt{x}}{\sqrt{5-x}+\sqrt{x}}dx$.

39. The probability density function of X is given by $f(x)=\begin{cases} ke^{-\frac{x}{3}} & \text{for } x>0 \\ 0 & \text{for } x\le 0 \end{cases}$

Find the value of k

40. Show that $y=e^{-x}+mx+n$ is a solution of the differential equation $e^x\left(\frac{d^2y}{dx^2}\right)-1=0$.

PART-IV

7x5=35

Answer all the questions:

41a) Investigate for what values of λ and μ

$x+2y+z=7,\ x+y+\lambda z=\mu, x+3y-5z=5$

has (i) no solution (ii) a unique solution (iii) an infinite number of solutions.

(OR)

b) Find the angle between $y=x^2$ and $y=(x-3)^2$.

42 a) If $z=x+iy$ and $\arg\left(\frac{z-i}{z+2}\right)=\frac{\pi}{4}$, show that $x^2+y^2+3x-3y+2=0$.

(OR)

b) Prove that $\int_0^{\frac{\pi}{4}}\log(1+\tan x)dx=\frac{\pi}{8}\log 2$.

43 a) Solve: $\cot^{-1}x-\cot^{-1}(x+2)=\frac{\pi}{12},\ x>0$.

(OR)

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b) On lighting a rocket cracker it gets projected in a parabolic path and reaches a maximum height of $4m$ when it is $6m$ away from the point of projection. Finally it reaches the ground $12m$ away from the starting point. Find the angle of projection.

44 a) Prove by vector method that $\sin(\alpha+\beta)=\sin\alpha\cos\beta+\cos\alpha\sin\beta$.

(OR)

b) A garden is to be laid out in a rectangular area and protected by wire fence. What is the largest possible area of the fenced garden with 40 metres of wire.

45 a) Find the centre, foci, and eccentricity of the hyperbola $11x^2-25y^2-44x+50y-256=0$

(OR)

b) If $u=\sin^{-1}\left(\frac{x+y}{\sqrt{x}+\sqrt{y}}\right)$, Show that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\frac{1}{2}\tan u$.

46a) Solve $\frac{dy}{dx}+\frac{3y}{x}=\frac{1}{x^2}$, given that $y=2$ when $x=1$

(OR)

b) Find the parametric form of vector equation, and Cartesian equations of the plane containing the line $\vec{r}=(\hat{i}-\hat{j}+3\hat{k})+t(2\hat{i}-\hat{j}+4\hat{k})$ and perpendicular to plane $\vec{r}\cdot(\hat{i}+2\hat{j}+\hat{k})=8$.

47 a) Prove that $p\rightarrow(\neg q\vee r)\equiv\neg p\vee(\neg q\vee r)$ using truth table.

(OR)

b) If X is the random variable with probability density function $f(x)$ given by,

$$f(x)=\begin{cases}x-1, & 1\le x<2\\ -x+3, & 2\le x<3\\ 0 & \text{otherwise}\end{cases}$$

find (i) the distribution function $F(x)$

(ii) $P(1.5\le X\le 2.5)$

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